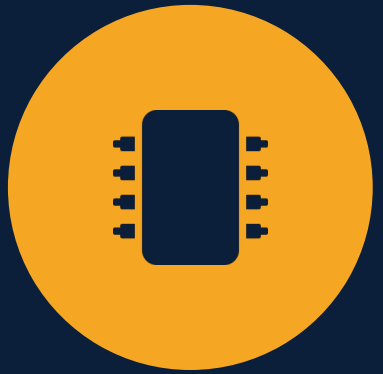
UNIT 3 · 9 hours

Zener Diode: Construction and Characteristics



Diodes, Transistors and their Applications

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Describe the Zener diode and its symbol
- Explain Zener and avalanche breakdown
- Interpret the V-I characteristic
- Compare Zener and ordinary PN diodes

Syllabus context

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

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Zener diode

Definition: heavily doped PN junction designed to operate in the reverse breakdown region.

Forward bias: behaves like an ordinary silicon diode.

Reverse bias: blocks until VZ, then conducts heavily while the voltage stays almost constant.

Symbol: diode with a bent cathode bar (Z shape).

VZ: set by doping level; commercial range ≈ 2.4 V to 200 V.

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Breakdown mechanisms

Zener breakdown (VZ < 5 V)

Heavy doping makes the depletion region very thin. A field of about 3×10^5 V/cm pulls electrons from covalent bonds. VZ decreases as temperature rises.

Avalanche breakdown (VZ > 8 V)

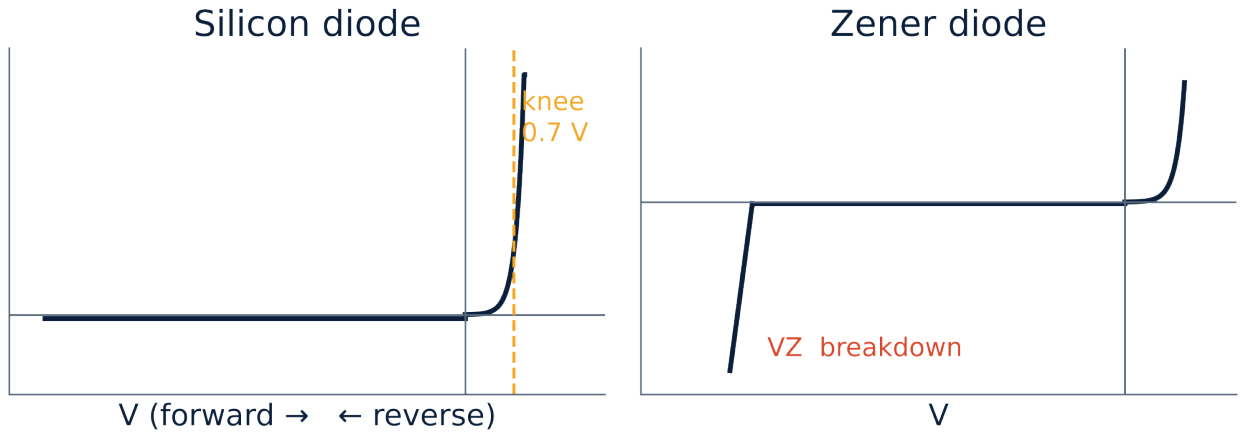
Minority carriers accelerate and collide with atoms, creating electron–hole pairs in a cumulative multiplication. VZ increases with temperature.

Between 5 V and 8 V

Both mechanisms act; temperature coefficient nearly zero.

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V-I characteristic



- Forward: like a silicon diode (0.7 V).
- Reverse: tiny leakage until VZ.
- Breakdown: current rises sharply; voltage nearly constant at VZ.
- Operate between IZmin (knee) and IZmax (power limit).

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Zener diode versus ordinary PN diode

Feature	PN diode	Zener diode
Doping	Light to moderate	Heavy
Depletion region	Wide	Narrow
Reverse operation	Avoided (damage)	Intended region of use
Breakdown	High, destructive	Sharp, reversible
Use	Rectification, switching	Regulation, reference

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Specifications and applications

Key ratings

VZ (nominal), IZmin, IZmax, power rating PZ = VZ IZmax (0.25 W–5 W), dynamic resistance rz.

Applications

Voltage regulation, reference voltage source, over-voltage protection, clipping and wave shaping.

Maximum current

IZmax = PZ / VZ, e.g. 2 W, 5 V → 0.4 A.

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Key takeaways

- Zener = heavily doped diode used in reverse breakdown
- Zener effect below ≈ 5 V; avalanche above ≈ 8 V
- Output voltage ≈ VZ over a wide current range
- A series resistor is always needed

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Practice questions

Q1. Explain Zener and avalanche breakdown.

Q2. A 6.8 V, 1 W Zener: find IZmax.

Q3. Compare a Zener and a PN diode.

Q4. Why is the Zener diode heavily doped?